

EX : 8

Producer and consumer using semaphores

Aim:

To implement the producer-consumer using semaphores

code:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int mutex = 1, full = 0, empty = 5, x = 0;
```

```
int wait(int s)
```

```
{ return --s;
```

```
}
```

```
int signal(int s)
```

```
{
```

```
return ++s;
```

```
}
```

```
void Producer()
```

```
{
```

```
mutex = wait(mutex);
```

```
empty = wait(empty);
```

```
full = signal(full);
```

```
x++;
```

```
printf("In producer produces item %d", x);
```

```
mutex = signal(mutex);
```

```
}
```


full = wait (full);

empty = signal (empty);

Printf ("In consumer item %d", x);

x --;

mutex = signal (mutex);

}

int main ()

{ int n;

Printf ("In Problem 1.");

Printf ("In Produce...");

Printf ("In consumer");

Printf ("In 3. Exit");

while (1)

{ Printf ("Enter your choice:");

scanf ("%d", &n);

switch (n)

{

case 1;

if (mutex == 1 && empty == 0)

{

Produce = 0;

}

else

Printf ("cannot produce");

}

break;

output:-

Producer item consumer problem

1. Producer

2. Consumer

3. Exit

Enter your choice : 1

* Producer produces item

Enter : 2

consumer ^{consumes} produces item

0 1 0
0 0 2
2 0 2
2 2 2
1 1 2
2 2 2
0 0 0

Result:-

Thus the Producer and consumer problem of semaphores.